Sharing the Experience of Awe: effects on Depth of Conversation and Interconnectedness

Preregistration

02/02/2026

TO ADD:

Measure for small self? Consider Tyson et al. 2022.

- Also, I created a pictorial "relational small self" item with JS on Qualtrics for another study, we could include it.

Complete adding all models used during analysis.

- ~~Add social anxiety scale and that it is adjusted for the context of the conversation (does not need to be done now, just mention it)~~
- MP:
  - Look for the connection scale
    - If we keep IOS, adjust scale description
  - Add authenticity and depth secondary features
  - Brms
  - Later:
    - Adjust scales
  - Exclusion criteria

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TO DO (NEW):

- Reorganize section 1 to include social anxiety. Include also understanding in paragraph 1.

1. Have any data been collected for this study already?

No.

2. What's the main question being asked or hypothesis being tested in this study?

The study will examine whether sharing an awe-inducing experience increases (a) self-reported depth of the conversation and (b) self-reported interpersonal interconnectedness with a conversation partner, compared to a neutral experience.

Hypotheses.

Hp1 – Awe leads to deeper conversations compared to the neutral condition.

Hp1a – Awe increases engagement with fundamental principles during the conversation (Depth Scale - Principles).

Hp2b – Awe increases the integrative complexity of the conversations (Depth Scale - Complexity).

Hp2 – Awe increases interpersonal interconnectedness with the conversation partner compared to the neutral condition.

Hp2a – Awe increases Inclusion of Other in the Self (IOS) scores.

Hp2b – Awe increases perceived understanding during the conversation.

3. Describe the key dependent variable(s) specifying how they will be measured.

Self-reported depth of conversation (Depth Scale; DS). Participants will complete a 14-item self-report scale immediately after each conversation. Items are answered on a 7-point Likert scale (1 = Strongly Disagree to 7 = Strongly Agree). An example item is: "During the reflection, many different thoughts were coming to my mind, covering distinct aspects of the issue." The scale is composed of two dimensions: engagement with fundamental principles (**DS-principles**, 7 items) and integrative complexity (**DS-complexity**, 7 items). The final scores will be the means of the 7-item subscales per condition; higher scores reflect greater perceived depth.

Commented [G(1]: compared to ...? And is having only a neutral condition as comparison enough, or should instead or in addition a positive condition be chosen?

Commented [RJ1R2]: Add "happiness" as control measure? - see is that explains differences in outcomes better than awe

Commented [G(2]: define once before abbreviating

Commented [MP3]: @everybody
What other social measures can we add that are related both to connectnedess and to awe?

Commented [RJ3R2]: - emotional syncrhony, though this is usually used for groups, we'd have to slightly adapt it to fit 2 people
- Positivity Resonance; characterized by shared positivity, mutual care and concern, and behavioral and biological synchrony. Measures exist for trait and episodic, would also need to be adapted

Commented [S(3R3]: check whether the conversation partners actually share feeling of awe explicitly when they discuss the view from the top floor.

Commented [S(3R4]: Another suggestion is to ask students to make and log observations about the interactions. Shared eye-contact, spatial closeness, mimicry of body posture, frequency of emotional facial expressions, dynamics in voice volume, expression of emotions in audio recordings, rating non-verbal signs of affiliation.

Commented [G(4]: compared to ...?

Commented [G(5]: Interesting scale. Good that it has multiple items and covers different elements of depth.

Deleted: The scale also measures other dimensions such as insightfulness (7 items), existential (7 items), analytical (3 items), and intuitive (4 items). However, the focus is on the dimensions of fundamental principles and integrative complexity…
¶

Interpersonal interconnectedness (Inclusion of Other in the Self; IOS). After each conversation, participants will complete the Inclusion of Other in the Self task (Aron et al., 1992). The IOS is a single-item pictorial measure in which participants select the pair of circles that best represents their perceived overlap/connectedness with their partner. Responses are …

Perceived understanding (Feeling Heard Scale). The participants are presented with 7 items, which they answer on a 7-point Likert scale. The Feeling Heard Scale (Roos et al., 2023) measures whether the participant felt heard during a conversation. The scale focuses on a multidimensional set of factors important for feeling heard, such as voice, attention, empathy, respect, and common ground.

Social Anxiety (State Social Anxiety Scale). The participants are presented with 7 7-point Likert-scale items. The State Social Anxiety scale (Kashdan & Steger, 2006) measures the degree to which the participant felt socially anxious during the conversation. The scale is adapted so it refers to the conversation had during the experiment.

Manipulation check (State awe). After each conversation, participants will indicate how much awe they experienced on a 1–7 Likert scale. This measure will be used to assess whether the awe-inducing condition elicited higher state awe than the control condition.

Big Five Inventory-10. Participants will fill in a 10-item questionnaire answered on a scale from 1 to 7 (Rammstedt & John, 2006). The questionnaire is a shorter version of the Big Five questionnaire. The scale measures trait extraversion, agreeableness, conscientiousness, neuroticism, and openness to experience.

The Dispositional Positive Emotion Scale (trait awe). The participants will fill in only the sub-scale 'awe' of the Dispositional Positive Emotions Scale (Shiota et al.,2006) , which contains 6 items. Participants will indicate how much they identify with items on a 7-point Likert scale, indicating trait awe.

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Commented [MP8]: Note: even if the origianal is 1-5, we do 1-7 for consistency with the others.

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Commented [MP10]: Add source

Commented [MP11]: Also good to add other measures (see the Qualtrics manipulation check measures).

Jasper suggested adding interest, attention, and/or arousal. Can you find good measures for these?

Commented [RJ11R2]: these as control-variables, or as an alternative manipulation check?

Commented [RJ11R3]: manipulation check

Commented [RJ11R4]: 2 or 3 items with synonyms

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Commented [MP12R2]: Look for IPIP-10

Commented [G(13]: Oxford comma fan

Commented [MP14]: Add source (Shiota et al)

Additional variables. Demographic information (e.g., age, gender, education) will also be collected.

4. How many and which conditions will participants be assigned to?

Participants will take part in dyads (pairs). Each dyad will complete both conditions (within-subject design):

Awe condition. In the awe condition (treatment), the conversation takes place on the top floor of an 88-meter-tall building, already validated in previous studies as a method for eliciting awe.

Neutral condition. In the neutral condition (control), the conversation takes place on the first floor of the same building, intended to be affectively neutral.

The order of conditions will be counterbalanced across dyads. In each condition, dyad members will engage in a 10-minute conversation based on a prompt introduced by the researcher, randomly selected from a list of 10 questions adapted from the Fast Friends / "36 questions" procedure (Aron et al., 1997; Sprecher, 2021). The questions asked during the conversations are:

1. Given the choice of anyone in the world, whom would you want as a dinner guest?
2. Would you like to be famous? In what way?
3. Before making a telephone call, do you ever rehearse what you are going to say? Why?
4. What would constitute a "perfect" day for you?
5. If you were able to live to the age of 90 and retain either the mind or body of a 30-year-old for the last 60 years of your life, which would you want?
6. For what in your life do you feel most grateful?
7. If you could wake up tomorrow having gained any one quality or ability, what would it be?

Commented [G(15]: Do you have a dedicated room for this free of distractions? I presume with first floor you mean one level up, so not on the ground floor where its busy, you have the cafetaria, and a lot of people meet.

Commented [S(15R2]: previously the ground floor was used if I am not mistaken. The ground floor is a similar open area as the top floor where many people walk by. It would be good nonetheless to include a measure of how busy it is, and perhaps also a measure of background noise which may be an indirect measure of distraction

Commented [MP16]: Have we tested if this is a reasonable duration?

Commented [MP16R2]: Should we allow participants to diverge from the conversation promt?

Commented [MP16R3]: TO DO: Specify

Commented [RJ16R4]: 1: need to try this out
2: yes, we allow

Commented [MP17]: We need to identify a subset of them appropriate for this study.

Commented [RJ17R2]: 36 questions are divided in 3 sets, each going further. The first set of 12 seems good for our purpose - they seem about even in depth, and not too intense to start talking about with a stranger.

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Commented [G(18]: Great

8. Is there something that you've dreamed of doing for a long time? Why haven't you done it?
9. What is your most treasured memory?
10. What is the greatest accomplishment of your life?

The questions will be counterbalanced. Each participant pair will be exposed to two different questions, one per condition. Participants are free to diverge from the given conversation topic, though they will be instructed to try to answer the question. This conversation will be audio-recorded using 2 clip-on microphones connected to 2 iPods. After each conversation, both participants will complete the measures for the dependent variables. The procedure will be repeated with the same partner in the second condition.

5. Specify exactly which analyses you will conduct to examine the main question/hypothesis.

Operationalization of the hypotheses

Hp1: Scores on the self-reported Depth Scale will be higher for participants having had a conversation on the 20th floor of the Erasmus building compared to those same participants

Hp2: Scores for self-reported interconnectedness will be higher for participants having had a conversation on the 20th floor of the Erasmus building compared to those same participants having had a conversation on the first floor of the Erasmus building.

Confirmatory analyses

All confirmatory tests comparing the awe and neutral conditions will employ Bayesian mixed-effects models to account for (a) the within-subject design (two observations per participant), (b) non-independence within given conversation topics, and (c) non-independence between partners within dyads.

For each of the dependent variables corresponding to our research hypotheses, will fit linear mixed-effects models of the following form:

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```
DV ~ condition
     + (1 + condition | dyad_ID/participant_ID)
     + (1 + condition | topic_ID)
```

participant_ID is the unique participant identifier, and dyad_ID identifies the pair in which each participant is nested.

To test potential moderators, we will extend the models as follows:

```
DV ~ condition * moderator
     + (1 + condition * moderator | dyad_ID/participant_ID)
     + (1 + condition * moderator | topic_ID)
```

Bayesian mixed-effects models will be estimated in R using brms::brm (…

…

In the event of the model not correctly handling data double-nested in participant_id and dyad_id, a simplified model without random effects for dyad_id will be used, of the form:

```
DV ~ condition + (1 + condition | participant_ID)
     + (1 + condition | topic_ID)
```

Checking assumptions

For linear mixed-effects models, we will check residual normality (visual inspection and Q–Q plot; Shapiro–Wilk test if needed), homoscedasticity, and linearity (residuals vs. fitted values). For generalized mixed models, we will check overdispersion and model fit diagnostics appropriate to the chosen distribution. If assumptions are

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Commented [MP22]: This does not necessarily make sense (if the moderator is an individual difference measured only once)

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Deleted: for linear mixed models), with Restricted Maximum Likelihood (REML), Kenward–Roger degrees of freedom approximation (method = "KR"), and Type III sum of squares (type = 3).¶

Deleted: Model fitting issues will be handled as follows. In case of singular fits, we will simplify the random-effects structure in a principled sequence (e.g., remove the participant-level random slope, then consider removing the dyad intercept if necessary) while keeping the fixed-effects structure unchanged. In case of convergence warnings, we will (1) check model specification and data coding; (2) standardize variables where appropriate; and (3) try alternative optimizers. If frequentist results are unstable, we will fit a Bayesian model using brms with weakly informative priors as a robustness check and compare inference across approaches.…

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Commented [MP23]: For myself: how do we specify this?

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violated, we will consider transformations of the dependent variable, robust standard errors, or a distributionally appropriate generalized model.

6. Any secondary analyses?

We will compute correlations between the self-reported repeated-measure outcomes (depth questionnaire, IOS, felt understood, social anxiety, and state awe) within each condition. These analyses are exploratory and intended to describe the relationships between the dependent variables.

MP: brms

As exploratory checks, we will examine whether pre-existing connection to the partner, similar or opposite gender to the partner, Big 5 personality scores, order of conditions, and prior visits to the top floor are associated with the primary outcomes and/or interact with condition, using the mixed-effects moderation models described in Section 5.

To examine whether the effect of awe is correlated across the self-reported outcomes (depth questionnaire, interconnectedness), we will follow up the main analysis with a Bayesian multivariate linear model of the following form:

Mvbf (DV1, DV2) ~ condition + (1 | p | participant_id / dyad_id) + (1 | q | topic_id)

MP

Analysis of the Conversations.

Do awe versus neutral conversations differ in transcript-based markers of conversational depth (frequency of depth-related themes/markers, normalized for transcript length)?

Do text segments representing depth as self-disclosure occur at more vulnerable levels and/or at levels involving more emotional risk in either of the conditions?

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Deleted: perceived depth and interpersonal connection

Commented [MK24]: This is ambiguous. If we go with a, say, male, female, non-binary approach, then in what way could there be "similarity of gender?

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Which specific depth-related themes/markers (derived from Cova et al., 2024) account for any observed condition differences?

Do transcript-based depth markers correlate with self-reported depth of conversation within and across conditions?

Do transcript-based depth markers correlate with interpersonal connectedness (IOS) and perceived understanding (felt understood)?

Do additional LLM-derived linguistic/semantic features (e.g., abstraction, topic complexity, emotional/existential content) converge with thematic coding and predict self-reported outcomes?

MP

Thematic Analysis (TA)

We will conduct a Thematic Analysis (TA) of the conversations recorded from participants. This allows explorative more qualitative investigation of depth and interconnectedness, as they emerge in awe and neutral conditions. This analysis will have both bottom-up and top-down parts, following a hybrid approach chosen to increase rigor (Fereday, J., & Muir-Cochrane, E. (2006). In the top-down part, … (1) Cova et al. (2024): *specify here, report in the appendix, and add three examples from hypothetical conversations* (2) degree of self-disclosure, with labels ordered based on the level of vulnerability of the topic (*Hein's list*) → appendix. … In the bottom-up part of the analysis … emergence of themes.

The procedure is outlined as follows:

1. Data cleaning, anonymization, and familiarization.
2. Depth-component based codes are applied, based on Cova et al (2024)
3. Use of Large Language Models (LLMs) to spot and map text segments that match the deductive codes . The LLM will receive the conversation text and will have to report …
4. Researcher's validation of LLM's matching process; subsequent calculation of inter-rater agreement between researcher and LLM, on the basis of a random selection of matched codes. We will include a second iteration of this procedure, specific to different levels of depth of conversational segments representing "depth as self-disclosure". The data will be ordinal. The scale by which self-disclosures are distinguished in depth ranges from shallower self-disclosures like disclosing basic public biographical facts (scored as 1) to deeper self-disclosures,

Commented [MK25]: two times "inter" seems once redundant

Commented [MP26]: We have to make this more specific (what to take as depth markers)

Commented [S(26R2]: It seems to me that depth in the context of the conversational task is mostly reflected in the sharing of authentic core beliefs. So perhaps we should look for self-disclosure as a measure of depth in different categories. Self-disclosure might be ordered by emotional risk and vulnerability:
Level 1 – Public/Biographical: Basic facts (name, age, job).
Level 2 – Preferences & Opinions: Tastes and general views (hobbies, music, political leaning).
Level 3 – Personality: Self-descriptions and traits (strengths, weaknesses, "I'm introverted").
Level 4 – Personal History: Formative experiences and life events (childhood memories, past relationships).
Level 5 – Core Values & Identity: Deep beliefs and identity-related experiences (moral views, identity struggles).
Level 6 – Emotional Vulnerabilities: Insecurities, fears, shame, loneliness.
Level 7 – Trauma & Deep Secrets: Highly sensitive experiences (abuse, severe mental health crises, profound loss).
Depth increases with personal significance, emotional exposure, and perceived risk.

Also self-disclosure could be reflected in emotional intensity in verbal and facial expression. Experimenters could rate conversations on the fly.

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Commented [MP27]: Update based on previous comment

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Commented [MP29]: Inter rater agreement:
Look for how it is computed statiscally and explain how.

Then, look specifically at the cases generated by LLMs only if there is a mismatch with human raters to assess possible reasons for the mismatch.

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that involve higher vulnerability and that carry significant interpersonal risk, such as disclosing a personal flaw (scored as 7).

5. In an additional step, emergent bottom-up codes are identified by LLM
6. Emergent codes are matched by LLM and researcher across a random sample of conversations to achieve inter rater reliability for emergent codes
   a. (7) Convergence onto a final map of themes
7. Comparison in frequency of occurrence of components of map between conditions
8. Correlation/Convergence with results obtained in quantitative self report scales.

In addition to this hybrid analysis, we will quantify four specific markers of conversational depth. These are "idea density" and "metaphors and analogical thinking".

(1) *Idea density*

How: We will measure the meaning density of words relative to the total number of words per conversation, that is, meaning in the sense of "semanticity".

Why: Idea density indicates a decrease in reliance on default everyday phrases, platitudes, but also a decreased reliance on phatic talk (meaning social- focused small talk that is not intended to convey meaning but to increase bonding (Oxford University Press, n.d.).

(2) *Metaphors and analogical thinking*.

How: Metaphorical and analogical thinking means comparing aspects of one thing with aspects of another thing, thereby crystallising/conveying a property they share.

Why: Measuring metaphors and analogical thinking indicates depth, because analogical thinking is evidence for having extracted and reapplied a core pattern or principle, and—to that extent—having achieved depth beyond obvious meaning extraction.

For the Cova et al. (…) thematic analysis, we will obtain … which we will analyze … .

For the self-disclosure thematic analysis, we will obtain … which we will analyze … .

For the … add all the others outputs …

Scaling of the conversations based on:
- Depth/profound vs. shallow/superficial
- Deep principles

Commented [MP31]: Add entire rubric to the Appendix

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Commented [MP34]: we need a statistical test for this

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Commented [MP35]: Use correct quatation marks across the text (curly)

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Commented [S(36]: unclear how to measure this?

Commented [MP36R2]:

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Commented [MP38]: Specify how we will do this (e.g., count the number of independent occurrences

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Commented [MP39]: ? is this a technical term taken from somewhere?

Commented [MK39R2]: No, I mean, thereby putting into sharp relief/ giving form to

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- Complexity: differentiation
- Complexity: integration

For exploratory purposes… (bottom-up things)

***Semantic Network Analysis* (SNA; Segev, 2021)**

SNA will be conducted on each conversation to capture the differentiation and integration aspects of depth by looking at the amount of unique words and distance between these words. A higher amount of concepts that appear in a conversation will indicate greater differentiation, while a shorter average path from one concept to another will indicate greater integration.

Stop-words will be removed from the transcripts (e.g., "the", "ehm", "it"), then words with similar meanings (e.g., plurals, different spellings...) will be merged into semantic-based nodes (i.e. "warmer", "climate", "intuition", "bathroom"). Nodes will be linked following the *k*-next neighbourhood method with a window of 3 words; whenever two nodes appear within 3 words of each-other, they are linked together.

R-packages SemNetDictionaries, SemNetCleaner, and SemNet will be used to prepare and analyze the networks (Alexander & Yoed , 2021). Text cleaning will be done using the general SemNet dictionary.

Community network methods will be used to find the average number of links between nodes, indicating differentiation. Pathfinder Network methods will be used to find the average shortest path between two nodes, indicating integration. The average amount of links will be divided by the average shortest path for a measure of depth. These scores will be used as an alternative to self-reported depth in the mixed-effects model described in section 5. Additionally, this measure of depth will be compared with participant's self-reported depth-scores using the following linear mixed-effects model:

```
Sem_score ~ self_report + (1 + self_report | participant_ID)  +
(1 | topic_ID)
```

MP (nested in pairs)

Commented [MP40]: Say: based on the Depth Scale

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Integrative Complexity Analysis (ICA, Smith et al., 1992)

Integrative Complexity Analysis will be applied to the conversations to capture the participants evaluative differentiation and conceptual integration, reflecting the "complexity" component of depth as defined by the Depth Scale.

Python and LiteLLM (a package that allows using the same script structure for different LLMs) will be used to train LLMs (e.g., Claude, Gemini, Llama) on the testing material of the *Motivation and Personality: Handbook of Thematic Content Analysis; Conceptual/Integrative Complexity* (Smith et al., 1992). Complexity integration focuses on the structure of thought in a text by looking at the conceptual structure of the conversation and not on the arguments given. Specifically, complexity integration **looks at** the amount of differentiation and/or integration present in a thought. Based on the amount of differentiation and/or integration a score from 1-7 is given to every thought. In case of this experiment a 'thought' will be a segment of conversation where only one participant speaks on one idea. Level 1 shows no evidence of either differentiation or integration. Level 3 indicates moderate to high levels of differentiation, but not integration. Level 5 shows moderate to higher differentiation and moderate integration. Level 7 indicates both high differentiation and integration. Levels 2, 4 and 6 are considered transitional levels, they contain an amount of differentiation and integration that lies between two levels.

Commented [RJ41]: Are we allowed to use this without a license or buying it or something?

As stated before, LLM will be used to apply this method to determine complexity of thought. This decision is made due to the long practicing time to manually learn to apply this method.

LLMs will (1) be asked to follow the Smith et al.'s (1992) guidelines, then tested on training material, and (2) trained on the training material without the pre-given method instructions. The resulting two models will be used to assess the conversations and assign a 1–7 score of Integrative Complexity.

These scores will be used as an alternative to self-reported depth in the mixed-effects model described in section 5. Since we will have scores from different LLMs for

each conversation, we will include random intercept and random slopes for each specific conversation and for the each LLM used.

… model …

(1 + … | conversation_ID) + (1 + … | LLM)

Additionally, this measure of depth will be compared with participant's self-reported depth-scores using the following linear mixed-effects model:

```
IC_score ~ self_report + (1 + self_report | participant_ID) + (1 | topic_ID) + (1 + self_report | conversation_ID) + (1 + self_report | LLM)
```

Pitch Analysis

An analysis of pitch will be used to identify how vocal dynamics shift during deep conversation and the interconnectedness between speakers. Audio recordings of each conversation will be uploaded into Praat Version 6.4.43 (Boersma & Weenink, 2025) to find the fundamental frequency (F_0; i.e., pitch) of each speaker. To create annotations (define) of each speaker segment, Python will be used to convert the transcripts with speaker labels into TextGrids compatible with Praat. Each TextGrid will be used to match each speaker with their F_0 and generate a time series for each speaker. Default pitch settings in Praat are between 75 and 600 Hz and will be adjusted based on the pitch range of each speaker. To analyze vocal synchrony, each parallel time series will be compared in R using Cross-Recurrence Quantification Analysis (CRQA) to calculate the coupling between speakers.

Commented [MK42]: This sentence is still too ambiguous

Commented [MK42R2]: adjusted "based on the individual pitch range of the speaker"

If CRQA is a single value:

1. Group differences in CRQA
2. DVs (self reported) ~ CRQA

If CRQA is NOT a single value:

1. Divided conversations based on the level of depth and connection (low, medium, high tertile based on the average score of the two participants taking part in each conversation)
2. Compare the CRQA patterns

TO ADD:

Extract other vocal features.

Regression:

Depth_scale_score ~ vocal_feature_1 + vocal_feature_2 + vocal_feature_3 … + (1 + … | …)

Outliers and Exclusions. Describe exactly how outliers will be defined and handled, and your precise rule(s) for excluding observations.

Outliers will be identified using the Median Absolute Deviation (MAD; Leys et al., 2013). As Leys et al. (2013) recommend, scores exceeding 3 MAD units will be identified as outliers and winsorized. This entails rescaling outliers by reassigning their values to the nearest value that lies within the valid (non-outlying) distribution. By doing this, accuracy and power are maximized, as all data points are maintained while the influence of outliers is reduced.

Commented [S(43]: this is probably a leftover from a previous study

Severe skewness ($|\gamma_1| \geq 1$) will be preferentially dealt with log e transformation.

8. How many observations will be collected or what will determine the sample size? No need to justify decision, but be precise about exactly how the number will be determined.

We aim at a sample size of 166 participants (83 dyads) with complete data for both conditions. Regardless of the sample size reached, data collection will end on June 2nd.

Commented [MP44]: Add an "hard deadline" for end of data collection.

9. Anything else you would like to pre-register? (e.g., data exclusions, variables collected for exploratory purposes, unusual analyses planned?)

No.

References

Alexander P C, Yoed N K (2021). "Semantic network analysis (SemNA): A tutorial on preprocessing, estimating, and analyzing semantic networks." *Psychological Methods*. doi:10.1037/met0000463

Aron, A., Aron, E. N., & Smollan, D. (1992). Inclusion of Other in the Self Scale and the structure of interpersonal closeness. *Journal of Personality and Social Psychology*, *63*(4), 596–612. https://doi.org/10.1037/0022-3514.63.4.596

Aron, A., Melinat, E., Aron, E. N., Vallone, R. D., & Bator, R. J. (1997). The experimental generation of interpersonal closeness: A procedure and some preliminary findings. *Personality and Social Psychology Bulletin*, *23*(4), 363–377. https://doi.org/10.1177/0146167297234003

Boersma, P., & Weenink, D. (2025). Praat: doing phonetics by computer (Version 6.4.43) [Computer program]. http://www.praat.org/

Cova, F., Schöpfer, C., & Bezat, M. M. (2024). Delving into depth: An empirical investigation of the ordinary concepts of depth and profundity. *Synthese*, *204*(5), 145. https://doi.org/10.1007/s11229-024-04786-7

Kashdan, T. B., & Steger, M. F. (2006). Expanding the topography of social anxiety. *Psychological Science*, *17*(2), 120–128. https://doi.org/10.1111/j.1467-9280.2006.01674.x

Leys, C., Ley, C., Klein, O., Bernard, P., & Licata, L. (2013). Detecting outliers: Do not use standard deviation around the mean, use absolute deviation around the median. *Journal of Experimental Social Psychology*, *49*(4), 764–766. https://doi.org/10.1016/j.jesp.2013.03.013

Naeem, M., Ozuem, W., Howell, K., & Ranfagni, S. (2023). A step-by-step process of thematic analysis to develop a conceptual model in qualitative research. *International Journal of Qualitative Methods*, *22*, 16094069231205789. https://doi.org/10.1177/16094069231205789

Oxford University Press. (n.d.). Phatic. In Oxford learner's dictionaries. Retrieved [06.03.2026], from oxfordlearnersdictionaries.com.

Rammstedt, B. & John, O. P. (2007). Measuring personality in one minute or less: A 10 item short version of the Big Five Inventory in English and German. Journal of Research in Personality, 41, 203-212.

Roos, C. A., Postmes, T., & Koudenburg, N. (2023). Feeling heard: Operationalizing a key concept for social relations. *PLoS ONE*, *18*(11), e0292865. https://doi.org/10.1371/journal.pone.0292865

Segev, E. (2021). Semantic network analysis in Social Sciences. https://doi.org/10.4324/9781003120100

Schoenherr, D., Strauss, B., Stangier, U., & Altmann, U. (2021). The influence of vocal synchrony on outcome and attachment anxiety/avoidance in treatments of

social anxiety disorder. Psychotherapy, 58(4), 510–522. https://doi.org/10.1037/pst0000393

Shiota, M. N., Keltner, D., & John, O. P. (2006). Positive emotion dispositions differentially associated with Big Five personality and attachment style. The Journal of Positive Psychology, 1(2), 61-71. https://doi.org/10.1080/17439760500510833

Smith, C. P., Atkinson, J. W., McClelland, D. C., & Veroff, J. (Eds). (1992). *Motivation and personality: Handbook of thematic content analysis* (pp. xvi, 708). Cambridge University Press. https://doi.org/10.1017/CBO9780511527937

Sprecher, S. (2021). Closeness and other affiliative outcomes generated from the Fast Friends procedure: A comparison with a small-talk task and unstructured self-disclosure and the moderating role of mode of communication. *Journal of Social and Personal Relationships*, *38*(5), 1452–1471. https://doi.org/10.1177/0265407521996055